

Chapitre 1 :

Atome, élément & tableau périodique

1. Constitution de l'atome
- 2. Les éléments et le tableau périodique**
3. La configuration électronique et la valence

Élément : substance chimique élémentaire, pure et indissociable.

- Désigné par un nom, qui dépend de la langue

FR : oxygène

EN : oxygen

DE : Sauerstoff

- Désigné par symbole chimique qui a une validité internationale
 - conçu par le chimiste suédois Jöns Jacob Berzelius (1779-1848)
 - une lettre majuscule, parfois suivie d'une lettre minuscule selon le nom latin

<i>Oxygenium</i>	O	oxygène
<i>Carbonium</i>	C	carbone
<i>Nitrogenum</i>	N	Azote
<i>Natrium</i>	Na	sodium

alcalins

alcalino terreux

Coll6: chalcogènes

Coll7: halogènes

Coll8: gaz nobles

Périodes

IUPAC Periodic Table of the Elements

Key:

atomic number	Symbol	name	conventional atomic weight	standard atomic weight
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métaux de transition

1 H hydrogen 1.008 [1.0078, 1.0082]	2 He helium 4.0026																
3 Li lithium 6.94 [6.938, 6.987]	4 Be beryllium 9.0122																
11 Na sodium 22.990	12 Mg magnesium 24.305 [24.304, 24.307]																
19 K potassium 39.098	20 Ca calcium 40.078(4)	21 Sc scandium 44.956	22 Ti titanium 47.867	23 V vanadium 50.942	24 Cr chromium 51.996	25 Mn manganese 54.938	26 Fe iron 55.845(2)	27 Co cobalt 58.933	28 Ni nickel 58.693	29 Cu copper 63.546(3)	30 Zn zinc 65.38(2)	31 Ga gallium 69.723	32 Ge germanium 72.630(8)	33 As arsenic 74.922	34 Se selenium 78.971(8)	35 Br bromine 79.904 [79.901, 79.907]	36 Kr krypton 83.798(2)
37 Rb rubidium 85.468	38 Sr strontium 87.62	39 Y yttrium 88.906	40 Zr zirconium 91.224(2)	41 Nb niobium 92.906	42 Mo molybdenum 95.95	43 Tc technetium [98.01, 98.02]	44 Ru ruthenium 101.07(2)	45 Rh rhodium 102.91	46 Pd palladium 106.42	47 Ag silver 107.87	48 Cd cadmium 112.41	49 In indium 114.82	50 Sn tin 118.71	51 Sb antimony 121.76	52 Te tellurium 127.60(3)	53 I iodine 126.90	54 Xe xenon 131.29
55 Cs caesium 132.91	56 Ba barium 137.33	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59	81 Tl thallium 204.38 [204.38, 204.39]	82 Pb lead 207.2	83 Bi bismuth 208.98	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson



57 La lanthanum 138.91	58 Ce cerium 140.12	59 Pr praseodymium 140.91	60 Nd neodymium 144.24	61 Pm promethium	62 Sm samarium 150.36(2)	63 Eu europium 151.96	64 Gd gadolinium 157.25(3)	65 Tb terbium 158.93	66 Dy dysprosium 162.50	67 Ho holmium 164.93	68 Er erbium 167.26	69 Tm thulium 168.93	70 Yb ytterbium 173.05	71 Lu lutetium 174.97
89 Ac actinium	90 Th thorium 232.04	91 Pa protactinium 231.04	92 U uranium 238.03	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

lanthanides

actinides

For notes and updates to this table, see www.iupac.org. This version is dated 1 December 2018.
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ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ

ОСНОВАННОЙ НА ИХЪ АТОМНОМЪ ВѢСѢ И ХИМИЧЕСКОМЪ СХОДСТВѢ

			Tl = 50	Zr = 90	? = 180.
			V = 51	Nb = 94	Ta = 182
			Cr = 52	Mo = 96	W = 186.
			Mn = 55	Rh = 104,4	Pt = 197,4.
			Fe = 56	Ru = 104,4	Ir = 198
			Ni = 59	Co = 59	Pt = 106,6 Os = 199.
			Cu = 63,4	Ag = 108	Hg = 200
H = 1	Be = 9,4	Mg = 24	Zn = 65,2	Cd = 112	
	B = 11	Al = 27,4	? = 68	Ur = 116	Au = 197?
	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?
	O = 16	S = 32	Se = 79,4	Te = 128?	
	F = 19	Cl = 35	Br = 80	I = 127	
	Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133
			Ca = 40	Sr = 87,6	Ba = 137
			? = 45	Ce = 92	Pb = 207
			?Er = 56	La = 94	
			?Yt = 60	Di = 95	
			?In = 75,6	Th = 118?	
					Tl = 204

Д. Менделѣевъ

Les métaux alcalins :

- X : lithium, sodium, potassium, rubidium, césium
- Forment des oxydes X_2O et des composés ioniques avec le chlore XCl
- Réagissent avec l'eau



Dimitri Ivanovitch
Mendeleïev
1834 - 1907



ОПЫТЪ СИСТЕМЫ ЭЛЕМЕНТОВЪ

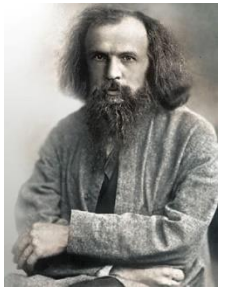
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	C = 12	Si = 28	? = 70	Sn = 118	
	N = 14	P = 31	As = 75	Sb = 122	Bi = 210?
	O = 16	S = 32	Se = 79,4	Te = 128?	
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Li = 7	Na = 23	K = 39	Rb = 85,4	Cs = 133	Tl = 204
	Ca = 40	Sr = 87,6	Ba = 137	Pb = 207	
	? = 45	Ce = 92			
	?Er = 56	La = 94			
	?Yt = 60	Di = 95			
	?In = 75,6	Th = 118?			

Д. Менделѣевъ

Les halogènes :

- X : fluor, chlore, brome, iode
- peuvent se combiner avec des métaux pour former de sels



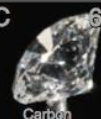
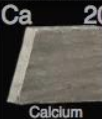
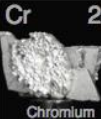
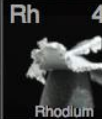
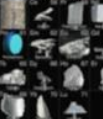
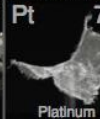
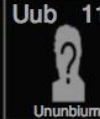
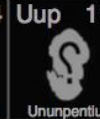
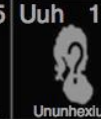
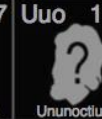
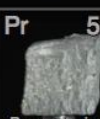
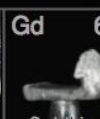
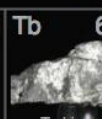
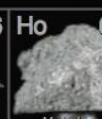
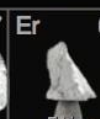
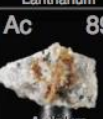
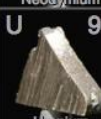
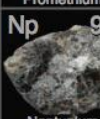
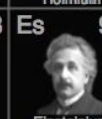
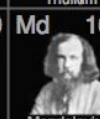
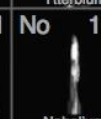
Dimitri Ivanovitch
Mendeleïev
1834 - 1907



Forment des halogénures HX, par réaction du corps pur simple X_2 avec l'hydrogène moléculaire H_2 .

The Elements

The Elements

 H 1 Hydrogen																	 He 2 Helium						
 Li 3 Lithium	 Be 4 Beryllium																	 B 5 Boron	 C 6 Carbon	 N 7 Nitrogen	 O 8 Oxygen	 F 9 Fluorine	 Ne 10 Neon
 Na 11 Sodium	 Mg 12 Magnesium																	 Al 13 Aluminum	 Si 14 Silicon	 P 15 Phosphorus	 S 16 Sulfur	 Cl 17 Chlorine	 Ar 18 Argon
 K 19 Potassium	 Ca 20 Calcium	 Sc 21 Scandium	 Ti 22 Titanium	 V 23 Vanadium	 Cr 24 Chromium	 Mn 25 Manganese	 Fe 26 Iron	 Co 27 Cobalt	 Ni 28 Nickel	 Cu 29 Copper	 Zn 30 Zinc	 Ga 31 Gallium	 Ge 32 Germanium	 As 33 Arsenic	 Se 34 Selenium	 Br 35 Bromine	 Kr 36 Krypton						
 Rb 37 Rubidium	 Sr 38 Strontium	 Y 39 Yttrium	 Zr 40 Zirconium	 Nb 41 Niobium	 Mo 42 Molybdenum	 Tc 43 Technetium	 Ru 44 Ruthenium	 Rh 45 Rhodium	 Pd 46 Palladium	 Ag 47 Silver	 Cd 48 Cadmium	 In 49 Indium	 Sn 50 Tin	 Sb 51 Antimony	 Te 52 Tellurium	 I 53 Iodine	 Xe 54 Xenon						
 Cs 55 Cesium	 Ba 56 Barium			 Hf 72 Hafnium	 Ta 73 Tantalum	 W 74 Tungsten	 Re 75 Rhenium	 Os 76 Osmium	 Ir 77 Iridium	 Pt 78 Platinum	 Au 79 Gold	 Hg 80 Mercury	 Tl 81 Thallium	 Pb 82 Lead	 Bi 83 Bismuth	 Po 84 Polonium	 At 85 Astatine	 Rn 86 Radon					
 Fr 87 Francium	 Ra 88 Radium			 Rf 104	 Db 105	 Sg 106	 Bh 107	 Hs 108	 Mt 109	 Ds 110	 Rg 111	 Uub 112	 Uut 113	 Uuq 114	 Uup 115	 Uuh 116	 Uus 117	 Uuo 118					
periodictable.com © 2007 Theodore Gray		 La 57 Lanthanum	 Ce 58 Cerium	 Pr 59 Praseodymium	 Nd 60 Neodymium	 Pm 61 Promethium	 Sm 62 Samarium	 Eu 63 Europium	 Gd 64 Gadolinium	 Tb 65 Terbium	 Dy 66 Dysprosium	 Ho 67 Holmium	 Er 68 Erbium	 Tm 69 Thulium	 Yb 70 Ytterbium	 Lu 71 Lutetium							
 Ac 89 Actinium	 Th 90 Thorium	 Pa 91 Protactinium	 U 92 Uranium	 Np 93 Neptunium	 Pu 94 Plutonium	 Am 95 Americium	 Cm 96 Curium	 Bk 97 Berkelium	 Cf 98 Californium	 Es 99 Einsteinium	 Fm 100 Fermium	 Md 101 Mendelevium	 No 102 Nobelium	 Lr 103 Lawrencium									

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Quels éléments sont des métaux?

- brillants et malléables
- conduisent bien la chaleur et l'électricité.
- La plupart (sauf Hg) sont solides à température ambiante.
- Dans les réactions, tendance à former des ions positifs

1 H hydrogen 1.008 [1.0078, 1.0082]																								2 He helium 4.0026											
3 Li lithium 6.94 [6.938, 6.997]		4 Be beryllium 9.0122														5 B boron 10.81 [10.806, 10.821]		6 C carbon 12.011 [12.009, 12.012]		7 N nitrogen 14.007 [14.006, 14.008]		8 O oxygen 15.999 [15.999, 16.000]		9 F fluorine 18.998		10 Ne neon 20.180									
11 Na sodium 22.990		12 Mg magnesium 24.305 [24.304, 24.307]																13 Al aluminium 26.982		14 Si silicon 28.085 [28.084, 28.086]		15 P phosphorus 30.974		16 S sulfur 32.06 [32.059, 32.076]		17 Cl chlorine 35.45 [35.446, 35.457]		18 Ar argon 39.95 [39.792, 39.963]							
19 K potassium 39.098		20 Ca calcium 40.078(4)		21 Sc scandium 44.956		22 Ti titanium 47.867		23 V vanadium 50.942		24 Cr chromium 51.996		25 Mn manganese 54.938		26 Fe iron 55.845(2)		27 Co cobalt 58.933		28 Ni nickel 58.693		29 Cu copper 63.546(3)		30 Zn zinc 65.38(2)		31 Ga gallium 69.723		32 Ge germanium 72.630(8)		33 As arsenic 74.922		34 Se selenium 78.971(8)		35 Br bromine 79.904 [79.901, 79.907]		36 Kr krypton 83.798(2)	
37 Rb rubidium 85.468		38 Sr strontium 87.62		39 Y yttrium 88.906		40 Zr zirconium 91.224(2)		41 Nb niobium 92.905		42 Mo molybdenum 95.95		43 Tc technetium 101.07(2)		44 Ru ruthenium 101.07(2)		45 Rh rhodium 102.91		46 Pd palladium 106.42		47 Ag silver 107.87		48 Cd cadmium 112.41		49 In indium 114.82		50 Sn tin 118.71		51 Sb antimony 121.76		52 Te tellurium 127.60(3)		53 I iodine 126.90		54 Xe xenon 131.29	
55 Cs caesium 132.91		56 Ba barium 137.33		57-71 lanthanoids		72 Hf hafnium 178.49(2)		73 Ta tantalum 180.95		74 W tungsten 183.84		75 Re rhenium 186.21		76 Os osmium 190.23(3)		77 Ir iridium 192.22		78 Pt platinum 195.08		79 Au gold 196.97		80 Hg mercury 200.59		81 Tl thallium 204.38 [204.36, 204.39]		82 Pb lead 207.2		83 Bi bismuth 208.98		84 Po polonium		85 At astatine		86 Rn radon	
87 Fr francium		88 Ra radium		89-103 actinoids		104 Rf rutherfordium		105 Db dubnium		106 Sg seaborgium		107 Bh bohrium		108 Hs hassium		109 Mt meitnerium		110 Ds darmstadtium		111 Rg roentgenium		112 Cn copernicium		113 Nh nihonium		114 Fl flerovium		115 Mc moscovium		116 Lv livermorium		117 Ts tennessine		118 Og oganesson	

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1
1
H
hydrogen
1.008
[1.0078, 1.0082]

3
Li
lithium
6.94
[6.938, 6.997]

11	Na	sodium	22.990
----	----	--------	--------

19
K
potassium
39.098

Rb
rubidium
85.468

55	Cs	caesium	132.91
----	----	---------	--------

87	Fr	francium
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[illegible]

Quels éléments sont des non-métaux?

- Ont des propriétés plus variées que les métaux
- Conduisent mal la chaleur et l'électricité.
- La plupart sont gazeux, liquides ou solides à température ambiante.
- Dans les réactions, tendance à former des ions négatifs

1 H hydrogen 1.008 [1.0078, 1.0082]																	10 2 He helium 4.0026						
3 Li lithium 6.94 [6.938, 6.997]	4 Be beryllium 9.0122																	13 5 B boron 10.81 [10.806, 10.821]	14 6 C carbon 12.011 [12.009, 12.012]	15 7 N nitrogen 14.007 [14.006, 14.008]	16 8 O oxygen 15.999 [15.999, 16.000]	17 9 F fluorine 18.998	18 10 Ne neon 20.180
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L'hydrogène fait-il partie du groupe des alcalins?

L'homogénéité d'un groupes décroît quand son nombre augmente

The image illustrates the periodic table and the physical states of elements in Group 15. The periodic table shows the following elements highlighted:

- 1 H hydrogen
- 7 N nitrogen
- 15 P phosphorus
- 33 As arsenic
- 51 Sb antimony
- 83 Bi bismuth

Below the periodic table, a photograph shows the physical states of these elements:

- azote (nitrogen) in a beaker
- phosphore (phosphorus) as a red powder
- arsenic as dark crystals
- antimoine (antimony) as a silvery metal block
- bismuth as a silvery metal block

PURE AND APPLIED CHEMISTRY

Les gaz nobles, anciennement gaz inertes

Periodic table of elements showing noble gases (Group 18) and actinides (Period 7, elements 89-103).

A black silhouette of a bomb with a lit fuse, with a green arrow pointing from the noble gases and a red arrow pointing from the actinides.

A chest X-ray image with a yellow radiation warning symbol overlaid on the center.

Periodic table of elements showing noble gases (Group 18) and actinides (Period 7, elements 89-103).

INTERNATIONAL UNION OF PURE AND APPLIED CHEMISTRY

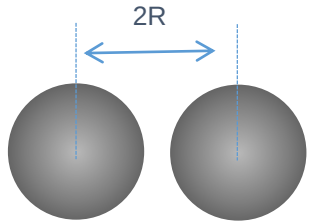
Comparons les propriétés des éléments, en fonction de leur position dans le tableau périodique

Quel atome est le plus petit?

Quel atome est le plus petit?																																			
1 H hydrogen 1.008 [1.0078, 1.0082]		2																13		14		15		16		17		18							
3 Li lithium 6.94 [6.938, 6.961]		4 Be beryllium 9.0122																5 B boron 10.81 [10.806, 10.821]		6 C carbon 12.011 [12.009, 12.012]		7 N nitrogen 14.007 [14.006, 14.008]		8 O oxygen 15.999 [15.999, 16.000]		9 F fluorine 18.998		10 Ne neon 20.180							
11 Na sodium 22.990		12 Mg magnesium 24.305 [24.304, 24.307]																13 Al aluminium 26.982		14 Si silicon 28.085 [28.084, 28.086]		15 P phosphorus 30.974		16 S sulfur 32.06 [32.059, 32.076]		17 Cl chlorine 35.45 [35.446, 35.457]		18 Ar argon 39.95 [39.942, 39.963]							
19 K potassium 39.098		20 Ca calcium 40.078(4)		21 Sc scandium 44.956		22 Ti titanium 47.867		23 V vanadium 50.942		24 Cr chromium 51.996		25 Mn manganese 54.938		26 Fe iron 55.845(2)		27 Co cobalt 58.933		28 Ni nickel 58.693		29 Cu copper 63.546(3)		30 Zn zinc 65.38(2)		31 Ga gallium 69.723		32 Ge germanium 72.630(8)		33 As arsenic 74.922		34 Se selenium 78.971(8)		35 Br bromine 79.904 [79.901, 79.907]		36 Kr krypton 83.798(2)	
37 Rb rubidium 85.468		38 Sr strontium 87.62		39 Y yttrium 88.906		40 Zr zirconium 91.224(2)		41 Nb niobium 92.906		42 Mo molybdenum 95.95		43 Tc technetium		44 Ru ruthenium 101.07(2)		45 Rh rhodium 102.91		46 Pd palladium 106.42		47 Ag silver 107.87		48 Cd cadmium 112.41		49 In indium 114.82		50 Sn tin 118.71		51 Sb antimony 121.76		52 Te tellurium 127.60(3)		53 I iodine 126.90		54 Xe xenon 131.29	
55 Cs caesium 132.91		56 Ba barium 137.33		57-71 lanthanoids		72 Hf hafnium 178.49(2)		73 Ta tantalum 180.95		74 W tungsten 183.84		75 Re rhenium 186.21		76 Os osmium 190.23(3)		77 Ir iridium 192.22		78 Pt platinum 195.08		79 Au gold 196.97		80 Hg mercury 200.59		81 Tl thallium 204.38 [204.38, 204.39]		82 Pb lead 207.2		83 Bi bismuth 208.98		84 Po polonium		85 At astatine		86 Rn radon	
87 Fr francium		88 Ra radium		89-103 actinoids		104 Rf rutherfordium		105 Db dubnium		106 Sg seaborgium		107 Bh bohrium		108 Hs hassium		109 Mt meitnerium		110 Ds darmstadtium		111 Rg roentgenium		112 Cn copernicium		113 Nh nihonium		114 Fl flerovium		115 Mc moscovium		116 Lv livermorium		117 Ts tennessine		118 Og oganesson	

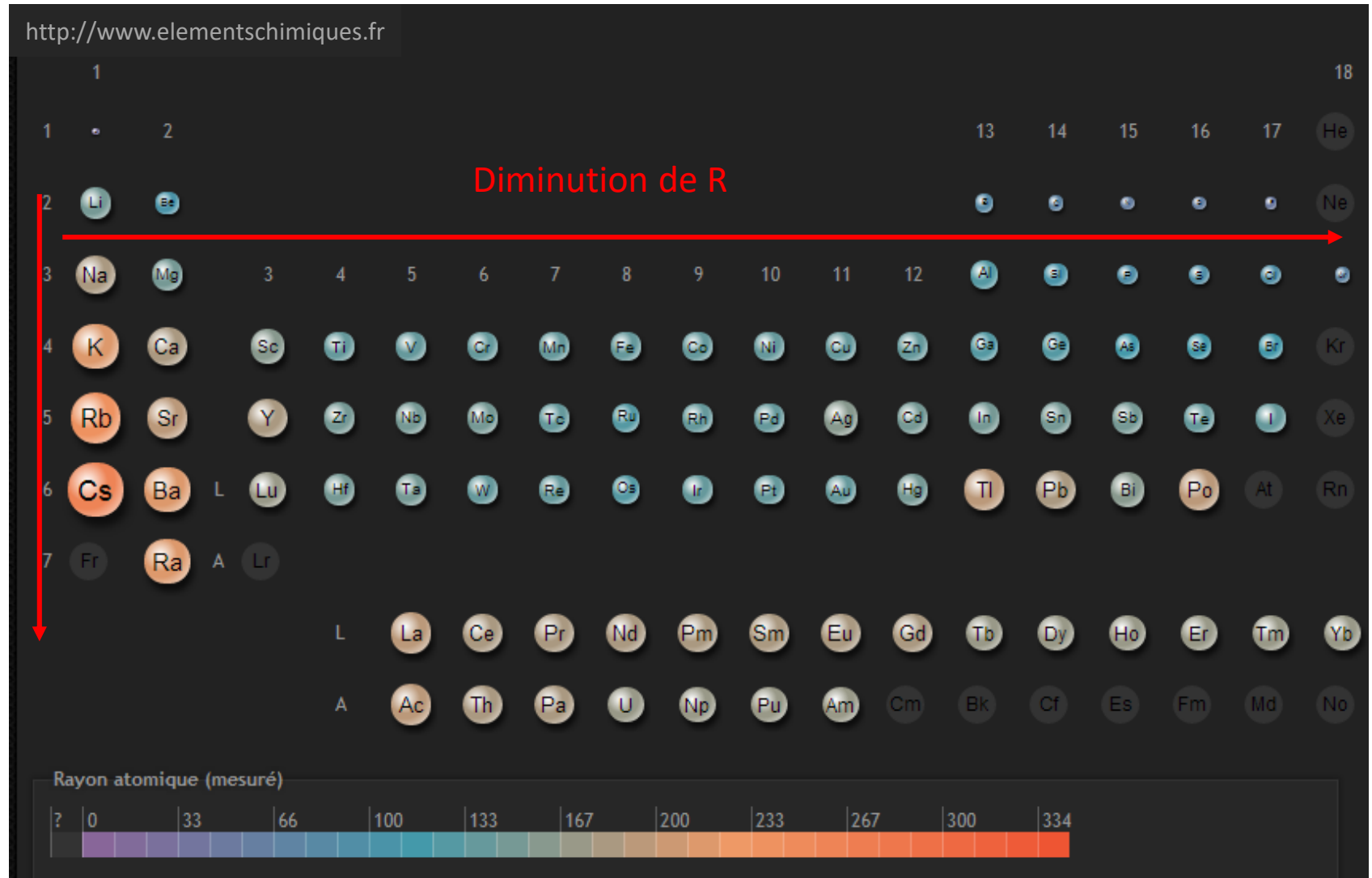
57 La lanthanum 138.91	58 Ce cerium 140.12	59 Pr praseodymium 140.91	60 Nd neodymium 144.24	61 Pm promethium	62 Sm samarium 150.36(2)	63 Eu europium 151.96	64 Gd gadolinium 157.25(3)	65 Tb terbium 158.93	66 Dy dysprosium 162.50	67 Ho holmium 164.93	68 Er erbium 167.26	69 Tm thulium 168.93	70 Yb ytterbium 173.05	71 Lu lutetium 174.97
89 Ac actinium	90 Th thorium 232.04	91 Pa protactinium 231.04	92 U uranium 238.03	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

Rayon atomique : moitié de la distance qui sépare 2 atomes d'un élément.

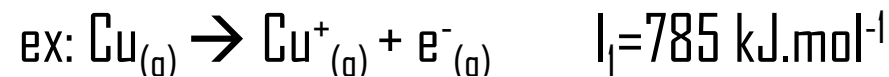
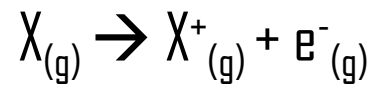


Le rayon est exprimé en pm
(picomètre = 10^{-12} m).

Augmentation de R



Energie de 1^{ère} d'ionisation : énergie requise pour arracher l'électron le moins lié à un atome en phase gazeuse.

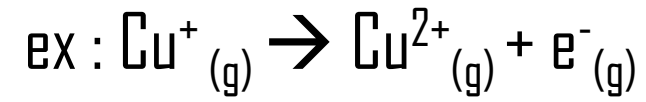


Augmentation de E_i

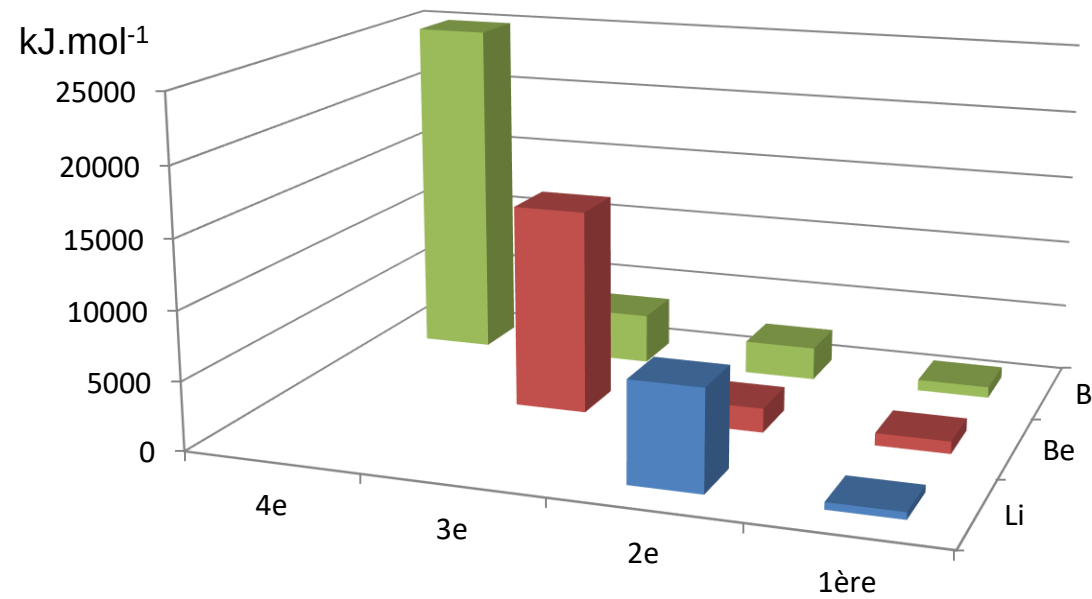
Diminution de E_i

1312																					2372					
520	899																				800	1086	1402	1314	1681	2080
495	738																				577	786	1012	999	1251	1520
419	589	3	4	5	6	7	8	9	10	11	12										579	762	944	941	1140	1351
403	549	39 Y yttrium 88.906	40 Zr zirconium 91.224(2)	41 Nb niobium 92.905	42 Mo molybdenum 95.95	43 Tc technetium	44 Ru ruthenium 101.07(2)	45 Rh rhodium 102.91	46 Pd palladium 106.42	47 Ag silver 107.87	48 Cd cadmium 112.41										558	708	830	869	1008	1170
375	503	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59										589	715	703	811		1037
392	509	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium										Valeurs données en kJ.mol ⁻¹					

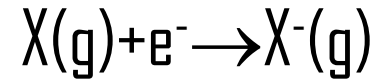
Energie de seconde ionisation : $X^+_{(g)} \rightarrow X^{2+}_{(g)} + e^-_{(g)}$



$I_1 = 1955 \text{ kJ.mol}^{-1}$



Affinité électronique : énergie émise par la capture d'un électron par un atome en phase gazeuse.



Variations moins régulières dans le TP, mais les halogènes groupe 17 et chalcogènes (groupe 16) ont les valeurs les plus fortes

	3	4	5	6	7	8	9	10	11	12
73	21 Sc scandium 44.956	22 Ti titanium 47.867	23 V vanadium 50.942	24 Cr chromium 51.996	25 Mn manganese 54.938	26 Fe iron 55.845(2)	27 Co cobalt 58.933	28 Ni nickel 58.693	29 Cu copper 63.546(3)	30 Zn zinc 65.38(2)
60	39 Y yttrium 88.906	40 Zr zirconium 91.224(2)	41 Nb niobium 92.906	42 Mo molybdène 95.95	43 Tc technetium 101.07(2)	44 Ru ruthenium 101.07(2)	45 Rh rhodium 102.91	46 Pd palladium 106.42	47 Ag silver 107.87	48 Cd cadmium 112.41
53	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59
48	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium
47	2	5								
46	14									

Ce qu'il faut retenir

- L'atome est la plus petite quantité de matière définissable pour un élément
- Au total, 118 éléments sont connus à ce jour, mais seuls 92 ont au moins un isotope stable.
- Le tableau périodique les classent par numéro atomique Z croissant, avec une organisation par groupe et par période.
- On distingue les éléments métalliques, les plus nombreux, et non métalliques.
- Les éléments ont des propriétés caractéristiques : taille, énergie d'ionisation, affinité électronique